DDD–B–421b
AUGUST 21, 1953

SUPERSEDING
Fed. Spec. DDD–B–421a
January 9, 1934

FEDERAL SPECIFICATION

BLANKETS; OTHER THAN ALL WOOL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers blankets of all cotton, cotton warp and wool filling, and blended nylon, wool, rayon, cotton and other fibers; primarily intended for use as bed covering. Class 2 blankets are also used as sheet and bath blankets.

1.2 Classification.

1.2.1 *Types and classes.*—Blankets covered by this specification shall be of the following types and classes, as specified:

Type I.—All cotton.
Class 1.—Twill (double filling).
Class 2.—Plain (single filling).

Type II.—Cotton-warp and wool or reprocessed wool filling.
Class 1.—Twill (double filling).

Type III.—Blended nylon-wool-rayon-cotton and other fibers.
Class 1.—Twill (double filling).

1.2.2 *Color.* — The color of the blankets shall be bleached white or dyed, as specified (see 3.8).

1.2.3 *Size.*—The blankets shall be of the sizes shown in table I and table II, as specified.

1.2.4 *Grade.*—Blankets shall be furnished in the grade known commercially as firsts.

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specifications of the issues in effect on date of invitation for bids form a part of this specification.

V–T–276—Thread; Cotton.
CCC–T–191—Textile Test Methods.
DDD–S–751—Stitches; Seams; and Stitching.

(Copies of Federal Specifications and the Index of Federal Specifications and Standards may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, SW., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Prices may be obtained from the Index of Federal Specifications and Standards or from the GSA Regional Offices. Single copies of this specification and other product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Missouri, Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification.

Rules and Regulations under the Wool Products Labeling Act of 1939.

(Copies of the Rules and Regulations under the Wool Products Labeling Act of 1939 may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price may be obtained from the Superintendent of Documents.)

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TABLE I.—*Physical requirements for type I*

Class	Size, minimum		Weight per blanket		Yarns per inch		Breaking strength, grab method	
	Width	Length	Maximum	Minimum	Warp, minimum	Filling, minimum	Warp, minimum	Filling, minimum
	Inches	*Inches*	*Pounds*	*Pounds*			*Pounds*	*Pounds*
1	60	84	3.2	3.0	37	36	30	25
1	66	84	3.5	3.3	37	36	30	25
1	66	90	3.7	3.4	37	36	30	25
2	70	84	1.3	1.2	35	24	28	15
2	72	90	1.7	1.5	27	28	27	17

2.3 Specifications and other publications applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.

3.1.1 *Cotton yarns.*—The cotton yarns shall be evenly spun of thoroughly cleaned cotton of a suitable grade and length of staple to meet the requirements for the finished blankets.

3.1.2 *Wool yarns.*—The wool yarns shall be of thoroughly cleaned fibers composed of virgin wool or reprocessed wool or any combination thereof, and shall be of a grade not lower than 44's U. S. Standard.

3.1.3 *Yarns of nylon-wool-rayon-cotton and other fibers.*—The blended fiber-yarns shall consist of the following: Not less than 80 percent of thoroughly cleaned wool fibers, not more than 10 percent of thoroughly cleaned rayon, cotton, and other fibers, and not less than 10 percent of thoroughly cleaned nylon fiber.

3.1.4 *Thread.*—The cotton thread for whip-stitching and binding shall be mercerized, type IC1, 3 cord, ticket No. B of Federal Specification V–T–276. The thread for stitching the satin-finished binding material shall be rayon, silk, or mercerized cotton and shall have a breaking strength of not less than 2 pounds (see 4.3.1).

3.1.5 *Satin (finish) binding material.*—When fabric binding is specified the fabric shall be of rayon or acetate, shall have a satin finish, and shall be not less than 4 inches wide, and shall weigh not less than 4 ounces per square yard. (See 4.3.2, 4.3.3, 4.3.4 and 4.3.5.)

3.2 Type I.—All cotton.

3.2.1 *Physical requirements.* — Type I blankets shall conform to the requirements shown in table I.

3.3 Type II.—Cotton warp and wool or reprocessed wool filling.

Type III.—Blended nylon-wool-rayon-cotton and other fibers.

3.3.1 *Physical requirements.*—Type II and type III shall conform to the requirements shown in table II.

3.4 Weave.

3.4.1 *Class 1.*—Class 1 blankets shall have a filling backed weave with one warp and two fillings and shall be any suitable twill weave. The blankets shall have selvage edges on the sides.

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TABLE II.—*Physical requirements for types II and III*

Class	Size, minimum		Weight per blanket		Yarns per inch		Breaking strength, grab method	
	Width	Length	Maximum	Minimum	Warp, minimum	Filling, minimum	Warp, minimum	Filling, minimum
	Inches	*Inches*	*Pounds*	*Pounds*			*Pounds*	*Pounds*
1	60	80	3.4	3.3	52	40	38	36
1	66	90	4.3	4.1	52	40	38	36
1	72	90	4.6	4.4	52	40	38	36
1	66	84	4.0	3.8	52	40	38	36

3.4.2 *Class 2.*—Class 2 blankets shall have a plain weave with one warp and one filling, woven with selvage edges on the sides.

3.5 Fiber content.

3.5.1 *Type I.*—The fiber content of type I blankets shall be 100 per cent cotton.

3.5.2 *Type II.*—The fiber content of type II blankets shall be not less than 80 percent by weight of virgin or reprocessed wool or any combination thereof, and not more than 20 percent cotton by weight (see 3.1.2 and 4.3.8).

3.5.3 *Type III.*—The fiber content of type III blankets shall be not less than 80 percent by weight of virgin or reprocessed wool or combination thereof, and not more than 10 percent rayon, cotton, and other fibers, and not less than 10 percent nylon fiber (see 3.1.3 and 4.3).

3.6 Napping.—The blankets shall be evenly napped on the face and back, and shall have a maximum of napping consistent with the other physical requirements.

3.7 Shrinkage.

3.7.1 *Type I.*—The shrinkage for type I blankets shall not exceed 10 percent in the warp direction and 5 percent in the filling direction when subjected to both laundering and dry cleaning (see 4.3.9 and 4.3.11).

3.7.2 *Types II and III.*—The shrinkage for types II and III blankets shall not exceed 10 percent in either the warp or filling direction when subjected to both laundering and dry cleaning (see 4.3.10 and 4.3.11).

3.8 Color.

3.8.1 *Blankets.* — The blankets shall be bleached white or dyed in pastel or dark colors, as specified. Whenever colors are specified the blankets shall show good colorfastness to laundering, crocking, dry cleaning, fresh and salt water, and light (see 4.3.12 through 4.3.17).

3.8.2 *Insignia.*—When stripes are specified they shall be woven. When insignia are desired they shall be stenciled, sewed, woven, or transfer marked, as specified. The colors shall show good colorfastness to laundering, dry cleaning, crocking, fresh and salt water, and light (see 4.3.12 through 4.3.17).

3.8.3 *Thread.* — The color of the sewing thread used for binding, whipstitching, and stitching the binding material shall blend with the color of the blanket, and shall show good colorfastness to laundering, crocking, dry cleaning, fresh and salt water, and light (see 4.3.18 through 4.3.24).

3.8.4 *Satin (finish) binding.*—The color of the satin-finished binding material shall blend with the color of the blanket, and shall show good colorfastness to laundering, crock-

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ing, fresh and salt water, and light (see 4.3.12 through 4.3.17).

3.9 Nonfibrous materials.—Sizing, finishing and other nonfibrous materials shall not exceed 5 percent (see 4.3.26).

3.10 Finished ends.—The cut ends of the blankets shall be finished by whipstitching, binding with fabric, or by turning under and hemming (binding) ½ inch, as specified.

3.11 Moth repellency.—When moth-repellent blankets are specified, the blankets shall be treated in such a manner as to be immune to the attack of insects such as clothes moths and carpet beetles when subjected to tests described in 4.3.30. The moth-repellent treatment shall be nonirritant and nontoxic and shall not alter the color or hand of the blanket.

3.12 Stitches, seams, and stitching.

3.12.1 *Whipstitching.*—Whipstitching shall be done with 14 ± 2 stitches per inch of stitch type 501, or other suitable stitch, as described in Federal Specification DDD–S–751. The stitching shall extend in from the edge at least ¼ inch.

3.12.2 *Binding stitching.* — Satin-finished binding material shall be sewn with 2 rows of stitching ¼ inch apart, using not less than 8 stitches per inch per row, of stitch type 301 as described in Federal Specification DDD–S–751. The ends of the binding shall have a liberal turn-under and shall be backstitched.

3.12.3 *Turned-end and hemmed stitching.*—The ends shall be turned under for ½ inch and hemmed and shall be sewn with 2 rows of stitching ¼ inch apart using not less than 8 stitches per inch per row, of stitch type 301 as described in Federal Specification DDD–S–751.

3.13 Wool label (types II and III only).—An appropriate label shall be securely attached to the blanket and shall contain the fiber content in accordance with the Wool Products Labeling Act of 1939 and the regulations authorized thereunder.

3.14 Workmanship.—The blankets shall be soft and full, evenly dyed, clean, and free from defects which affect the appearance or which may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Not less than one blanket of each type, class, size, and color shall be taken at random from each 1,000 blankets or fraction thereof offered for delivery, except that in no instance shall less than two blankets be taken from any shipment. When the shipment is over 10,000 blankets, one blanket shall be taken at random from each one-tenth of the shipment.

4.1.1 *Thread.*—Not less than 610 yards of each type, size, and color of sewing thread shall be submitted for test.

4.2 Inspection.—Inspection shall be made at the point of delivery, or at the factory, at the discretion of the purchasing agent.

4.3 Tests.—The tests shall be conducted in accordance with the applicable methods described in Federal Specification V–T–276 and the following standard conditions and methods described in CCC–T–191, and as hereinafter specified:

	Test	*Method No.*
	Standard condition	Section 4
4.3.1	Breaking strength (thread)	4100
4.3.2	Identification of acetate	1510
4.3.3	Identification of rayon:	
	Viscose	1500
	Cupprammonium	1520
4.3.4	Weight of cloth	5041
4.3.5	Width of cloth	5020
4.3.6	Yarns per inch	5050

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Test	Method No.
Standard condition	Section 4
4.3.7 Strength, breaking; grab method	5100
4.3.8 Wool content, alkali method	2101.1
4.3.9 Shrinkage in laundering; cotton and/or linen cloth	5550
4.3.10 Shrinkage in laundering; cloth other than cotton and linen	5552
4.3.11 Shrinkage in dry cleaning	5580
4.3.12 Colorfastness to laundering	5614
4.3.13 Colorfastness to dry cleaning	5620
4.3.14 Colorfastness to water	5630
4.3.15 Colorfastness to salt water	5632
4.3.16 Crocking of cloth (wet and dry)	5650
4.3.17 Colorfastness to light	5660
4.3.18 Colorfastness to laundering, cotton thread	4610
4.3.19 Colorfastness to laundering, rayon and silk thread	4614
4.3.20 Colorfastness to dry cleaning, thread	4620
4.3.21 Colorfastness to water, thread	4630
4.3.22 Colorfastness to salt water, thread	4632
4.3.23 Crocking resistance, thread	4650
4.3.24 Colorfastness to light, thread	4660
4.3.25 Nylon content of fiber mixtures	2530
4.3.26 Nonfibrous materials	2611.1
4.3.27 Length-weight relation, thread	4010
4.3.28 Colorfastness to chlorine bleaching, thread	4600
4.3.29 Identification of cotton	1200

4.3.30 *Moth repellent test.*—Three specimens each 12 x 12 inches shall be taken for test from each sample tested. One specimen shall be laundered 3 times, the second specimen shall be dry cleaned 3 times, and the third specimen shall be tested as received. Five discs each 3½ inches in diameter shall be taken from each of the above specimens, and shall be exposed to the destructive stage (larvae) of the buffalo moth, or the black carpet beetle, at a temperature of 68° to 83° F. under subdued light and other conditions favorable to larvae development and growth. After 30 days' exposure, there shall be no evidence of damage by the larvae.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Commercial packages will be acceptable under this specification.

5.2 Packing.—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking. — Shipping containers shall be marked with the name of the material, the type, the class, the color, the size and the quantity contained therein as defined by the contract or order under which shipment is made, the name of the contractor, and the number of the contract or order, the name of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data. — Purchasers should specify the type, class (see 1.2.1), color (see 1.2.2 and 3.8.1), and size (see 3.2.1 and 3.3.1). If any particular stripes, or insignia are desired, they should be specified (see 3.8.2). Purchasers should also exercise any desired options offered herein (see 3.10, 3.11, 4.2, and 6.2.).

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6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine conformance to the requirements of this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 **Transportation description.** — The transportation description applicable to these items is:

Blankets, made wholly of cotton, or mixed cotton and wool, the wool content not to exceed 50 percent by weight, in bales, boxes, or in rolls.

Blankets, made of a mixture of wool and rayon, the wool content not to exceed 50 percent by weight, in bales, boxes or wrapped rolls.

Carload weight, any quantity.

Blankets, N.O.I.

Truckload weight, any quantity.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standards of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army and Air Force.

7.2.1 *Applicable specifications and standards.*

Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood (for boxes whose weight of contents does not exceed 1,000 pounds).

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed (for weight of contents not in excess of 1,000 pounds).

MIL-B-107—Boxes, Wood, Wirebound (Military type).

JAN-P-125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

MIL-A-140 — Adhesive, Water-Resistant, Waterproof Barrier-Material.

MIL-B-3149—Baling of Clothing and Equipage.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by the contractor in connection with the specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Sampling, inspection, and test procedures.*

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7.2.2.1 *Sampling.*

7.2.2.1.1 *Army.* — Sampling shall be conducted in accordance with Military Standard MIL–STD–105.

7.2.2.2 *Inspection.*

7.2.2.2.1 *Army.*—Inspection shall be made by the Government to determine compliance with this specification.

7.2.3 *Packaging, packing, and marking.*

7.2.3.1 *Packaging.*

7.2.3.1.1 *Army and Air Force (Domestic and Overseas Shipment).* — Each blanket shall be folded to measure approximately 28½ by 14½ inches. Five folded blankets shall be neatly stacked in a bundle and securely tied with cotton twine or tape approximately 4 inches from each end.

7.2.3.2 *Packing.*

7.2.3.2.1 *Army (Domestic and Overseas Shipment).*—Blankets packaged as specified in 7.2.3.1.1 shall be packed in units of 20, in bales conforming to Military Specification MIL–B–3149, except that waterproof barrier material will not be required for domestic shipment.

7.2.3.2.2 *Air Force.* — Twenty blankets, packaged as specified in 7.2.3.1.1, shall be baled for domestic or overseas shipment in accordance with Military Specification MIL–B–3149.

7.2.3.2.2.1 When specified in the contract or order, the following alternate method of packing shall be used:

Boxes.—Twenty blankets, packaged as specified in 7.2.3.1.1 shall be packed in cleated plywood box, nailed wood box or a wirebound box designed for type 2 load and constructed in accordance with Military Specification JAN–P–105, JAN–P–106 or MIL–B–107, respectively. Closures and strapping shall comply with the appendix of the applicable box specification. Each box shall be lined with a sealed waterproof bag fabricated of material conforming to Military Specification JAN–P–125. Barrier shall be sealed with a waterproof adhesive conforming to the requirements of Military Specification MIL–A–140. A sufficient quantity of flaked moth repellent shall be evenly distributed throughout the pack. The moth repellent used shall not cause deterioration of the fabric and not cause bleeding or deterioration of any barrier material used.

7.2.3.3 *Marking.*

7.2.3.3.1 *Army.*—Shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

7.2.3.3.2 *Air Force.*—Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129. The nomenclature shall be as follows:

Blanket, Bed, Type ....* (Spell out applicable material)
Class....* (Spell out applicable data), Color ....*, Width....*, Length ....*
Design or Insignia (If applicable)

*Applicable data to be entered by the contractor.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

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